Hazardous Alcohol Consumption Effects on Depression and its Moderation by Gender

Bayley E. Camp

Student Study

Department of Psychology, University of Southern California

May 11, 2022

Abstract

This study investigated the relationship between hazardous alcohol consumption, depression, and gender. Specifically, we examined the impact of hazardous alcohol consumption on depression levels, and if males and females would have differing depression level increases based on their alcohol consumption. Participants (N = 114) answered both an Alcohol Use Disorders Identification Test (AUDIT) and Patient Health Questionnaire-9 (PHQ-9). We found that more than half of female participants had a score of moderately severe depression on the PHQ-9 and more than half of the male participants had a score of moderate depression on the PHQ-9. More than half of female and male participants scored as alcohol dependent on the AUDIT. Despite these findings, the results of our study indicate there was no statistical significance for gender moderating the relationship between hazardous alcohol consumption and depression.

Keywords: alcohol consumption, depression, gender

Hazardous Alcohol Consumption Effects on Depression and its Moderation by Gender

Hazardous alcohol consumption and depression have been studied for many years to determine if there is a causal relationship between the disorders. According to the 2019 NSDUH, "8.1% of adults ages 18 - 22 met criteria for past-year AUD. Within this age group, 8.7% were full-time college students" (U.S.Department of Health and Human Services, n.d.). The National Institute of Mental Health reported in 2020 that the prevalence of a major depressive episode in the previous year was highest (17.0%) among adults ages 18 – 25 (U.S.Department of Health and Human Services, n.d.). Caldwell, Rogers, (2002) found higher levels of anxiety and depression in male abstainers and males consumed alcohol in a hazardous manner. For females, they found only those who hazardously consumed alcohol had increased levels of depression and negative effect. Psychiatric disorders, such as anxiety and mood disorders, frequently co-occur with alcohol use disorder (AUD). Depressive disorders are the most common psychiatric disorders among people with AUD (Grant, Stinson, & Dawson, 2004). "AUD is frequently comorbid with other psychiatric disorders, and it is likely that AUD is both an etiologic factor in other mental disorders and a consequence of mental disease" (Flensborg-Madsen, 2008). "In the current issue, Boden & Fergusson report a meta-analysis that estimates a doubling of risk for alcohol use disorder (AUD) associated with major depression (MD) and, similarly, a doubling of risk for MD associated with AUD. The study is important, because it is based on epidemiological samples and suggests a 'moderate association' that cautions against overstating the relationship or dismissing it" (Connor, 2011). Geisner (2012) noted that while several studies found depressed

and non-depressed college students showed no differences in hazardous alcohol consumption, there are other research studies indicating individuals who experience depression are more likely to consume alcohol to aid with their problems. Studies such as these have established a complex, causal relationship between AUD and MD. Although this causal relationship has been established, it is important to conduct further research to determine if other factors increase depression or other psychiatric disorders with increased alcohol use. The present study investigated if gender influenced depression levels linked to hazardous alcohol consumption.

Past research regarding gender, depression, and hazardous alcohol consumption has shown a possible link to depression levels related to gender when it is impacted by excessive alcohol consumption. In a study of 15-year-old Finnish school students to determine social, psychological, and environmental factors related to heavy drinking, the authors determined that hazardous alcohol consumption "is associated with more severe psychosocial dysfunction" among females than males (Laukkanen, 2001). Another study determined a causal link between teenage problem behavior, including hazardous alcohol consumption, and adult psychopathology, including major depressive disorder. Twins were used for this study, each problem behavior included in the study was associated with clinical diagnosis, and showed a gender difference related to substance abuse and depressive disorder. "Among those with four or more problem behaviors before age 15, the lifetime rates of substance use disorders, antisocial personality disorder, and major depressive disorder exceeded 90%, 90%, and 30% in males and 60%, 35%, and 55% in females, respectively" (McGue & Iacono, 2005). Nolen-Hoeksema and Hilt (2010) found that hazardous alcohol use was more prevalent among men than women. The authors went

on to surmise that there may be compelling biological and social reasons that may be protective factors against AUDs in women, although they noted inconsistencies in the literature. They found this may be due to the fact that many studies do not have enough statistical power to detect gender differences. Similar complexities were found in a previous study by Nolen-Hoeksema and Harrell (2002) regarding rumination, depression, alcohol use and test of gender differences. While rumination was associated with depressive symptoms and a tendency to turn to alcohol in both men and women, it was a stronger predictor of alcohol-related problems in women, but not in men. In a population review in Norway, a review of 8983 young people determined "existing population based research also indicates important gender- and age differences in the prevalence of common adolescent mental health problems, with possible influence on the association with alcohol use" (Strandheim, 2012).

Although there are studies suggesting gender impacts depression associated with hazardous alcohol consumption, there are other studies that have not established a causal link.

Fergusson, Boden, and Horwood (2009) assessed causal links between AUD and MD. Their study included other factors such as gender to see if it had an effect. That 2009 study did not find evidence for gender effects in the association between AUD and MD, which they noted did not comport with the results of other studies. Boden and Fergusson (2011), in their later research on the association of AUD and MD stated, "The need for further research examining mechanisms of linkage, gender differences in associations between AUD and MD and classification issues was identified." This supports our reasoning for continuing this research.

The study being conducted is to determine the moderating effect of gender on depression related to hazardous alcohol consumption. This study will give insight into how alcohol consumption affects depression and how this differs by gender. The three variables of this study are: hazardous alcohol consumption, depression, and gender. We predicted a relationship between a person's alcohol consumption and depression and that gender would impact the correlation. We further expected that increased depression levels due to excessive alcohol consumption would be higher in males than females. Excessive alcohol consumption, or hazardous drinking, is a pattern of alcohol consumption that creates increased adverse health risk for people who partake in this risky behavior. This research will allow us to test our hypothesis that there is a relationship between a person's alcohol consumption and depression levels, and that it will be moderated by gender.

Method

Participants

Participants were recruited from Facebook, Instagram, Snapchat, and GroupMe. Our sample was from 114 participants who took a survey that was administered through Qualtrics. There were 98 females and 16 males. Participants were not required to have an existing depressive disorder diagnosis to participate in this study.

Materials

The materials used were computers, Qualtrics, and various social media platforms. Qualtrics is a service which allows you to create surveys. The survey was posted to various social media platforms (Facebook, Snapchat, Instagram, and GroupMe) to recruit participants. This allows the research to be conducted using convenience sampling. The participants were measured on three variables: alcohol consumption, depression, and gender. Alcohol consumption, a predictor variable, was measured using the Alcohol Use Disorders Identification Test (AUDIT). The outcome variable, depression, was measured using the Patient Health Questionnaire (PHQ-9).

AUDIT. AUDIT has become the most widely used alcohol screening since its publication in 1989. It is used to identify if a participant takes part in hazardous alcohol consumption. The AUDIT has 10 questions with questions one through eight being scored with 0, 1, 2, 3, or 4 (0 = *never to* 2 = 2-4 *times per month to* 4 = 4+ *times per week*), and questions nine and ten being scored with 0, 2, and 4. The range of possible scores is from 0 to 40 where the scores indicate the following:

0 indicates an abstainer who has never had any problems from alcohol.

1 to 7 suggests low-risk consumption.

8 to 14 suggest hazardous or harmful alcohol consumption.

15 or more indicates the likelihood of alcohol dependence (moderate-severe alcohol use disorder).

PHQ-9. The PHQ-9 test is widely used because of its high validity and reliability. Validity studies showed that those who scored a 10 or higher on the PHQ-9 were between 7% and 13.6%

more likely to be diagnosed with depression by a mental health professional. PHQ-9 score determines the severity of depression according to the diagnostic criteria from the Diagnostic and Statistical Manual of Mental Disorders. Participants answered nine questions related to their feelings and activities over the prior two weeks. These questions had a 4-point scale to assess symptoms of depression (0 = not at all to 4 = nearly every day). A PHQ-9 score total of 0-4 points equals “normal” or minimal depression. Scoring between 5-9 points indicates mild depression, 10-14 points indicates moderate depression, 15-19 points indicates moderately severe depression, and 20 or more points indicates severe depression.

In the first two questions, participants gave informed consent and provided demographic information regarding gender identity.

Results

We conducted a multiple linear regression analysis of the interaction between alcohol and depression for females and males. Our regression analysis of the relationship between alcohol consumption and depression levels determined that it is not moderated by gender because the beta coefficient which is (0.01159) does not have a statistically significant p-value. The p-value of 0.918 is greater than 0.05 (See Table 1). 3.903% of variability in alcohol consumption can be explained by gender and depression as indicated by the Multiple R-squared. This does not account for much of the variance in our model. The F-Statistic compares the joint effect of all variables together. The F-Statistic is 1.489 and the p-value of 0.2215 associated with the F-Statistic is greater than 0.05 indicating that our overall model is not showing statistical

significance. Our model does not support our hypothesis that gender moderates the relationship between hazardous alcohol consumption and Depressive Mood (DM).

Our data determined that many of our participants suffered from depression and engaged in hazardous alcohol consumption. The mean score for women on the PHQ-9 was 16.27, which is indicative of moderately severe depression (*see* Table 2 for the mean). The mean score for women on the AUDIT was 15.13, which indicates alcohol dependence (*see* Table 2 for the mean). Men scored a mean of 14.25 on the PHQ-9, which indicates moderate depression (*see* Table 2 for the mean). Men scored a mean of 16.25 on the AUDIT, which indicates alcohol dependence (*see* Table 2 for the mean). The Standard Deviation for PHQ-9 and the AUDIT for both men and women is not concentrated along the regression line in relation to the Mean, but scattered far and wide on either side of it (Figure 1).

Discussion

Our study results did not meet the criteria of statistical significance as indicated by the p-value of 0.918. This does not mean that our research study has provided concrete evidence to dispute a possible link between depression levels increased by hazardous drinking and a gender impact on those depression levels. These findings contribute to the mixed findings of other research studies regarding gender, depression, and alcohol. “For females, depressive symptoms predicted subsequent alcohol problems over 3 years (odds ratio = 3.04, 95% confidence interval 1.35–6.80, $p < 0.01$) and 4 years (odds ratio = 2.42, 95% confidence interval 1.14–5.12, $p < 0.05$), but not for 7 years. There was no evidence to support the hypothesis for males” (Steger Moscato, 1997). Zhan, Shaboltas, (2012) a significant correlation between alcohol use and depressive mood and

that interaction differed by gender. Men were more likely to report depressive symptoms if they scored in the first and fourth quartile of an AUDIT score while a linear trend was observed in women who consumed alcohol hazardously. Women were three times more likely to experience depressive symptoms if they abused alcohol than those who did not.

Our study was limited by small sample size. Also, this research study did not select only participants with a known depression diagnosis, nor did it have equal numbers of female and male participants. Research regarding the impact of hazardous alcohol consumption on depression levels and how it differs by gender is important. Our study did not show a statistical significance for gender moderating the relationship between hazardous alcohol consumption and increased depression levels. Future studies should continue to evaluate gender differences in relation to hazardous alcohol consumption, depression, and include other possible contributing factors. Understanding these differences for genders has the potential for providing better treatment protocols for AUD and depression in the future. “Additional research is needed to investigate relationships among alcohol use, depressive symptoms, gender differences, and other alcohol related risk factors in Russia. Preventive strategies for reduction of alcohol-related problems in Russia should consider these differences” (Zhan, Shaboltas, 2012).

References

Boden, J. M., & Fergusson, D. M. (2011). Alcohol and depression. *Addiction*, *106*(5), 906-914.

Caldwell, T. M., Rodgers, B., Jorm, A. F., Christensen, H., Jacomb, P. A., Korten, A. E., & Lynskey, M. T. (2002). Patterns of association between alcohol consumption and symptoms of depression and anxiety in young adults. *Addiction (Abingdon, England)*, *97*(5), 583–594. https://doi.org/10.1046/j.1360-0443.2002.00092.x

Geisner, I. M., Mallett, K., & Kilmer, J. R. (2012). An examination of depressive symptoms and drinking patterns in first year college students. *Issues in mental health nursing*, *33*(5), 280–287. https://doi.org/10.3109/01612840.2011.653036

Nolen-Hoeksema, S., & Hilt, L. (2006). Possible contributors to the gender differences in alcohol use and problems. *The Journal of general psychology*, *133*(4), 357–374. https://doi.org/10.3200/GENP.133.4.357-374

Nolen-Hoeksema, S., & Harrell, Z. A. (2002). Rumination, depression, and alcohol use: Tests of gender differences. *Journal of Cognitive Psychotherapy*, *16*(4), 391-403.

https://connect.springerpub.com/content/sgrjcp/16/4/391

U.S. Department of Health and Human Services. (n.d.). *Alcohol Facts and Statistics*. National Institute on Alcohol Abuse and Alcoholism. Retrieved April 27, 2022, from https://www.niaaa.nih.gov/publications/brochures-and-fact-sheets/alcohol-facts-and-statistics

U.S. Department of Health and Human Services. (n.d.). *Major depression*. National Institute of Mental Health. Retrieved April 27, 2022, from https://www.nimh.nih.gov/health/statistics/major-depression

Tables

Table 1

Statistical Results for multiple regression analysis

	Coefficient	Standard Error	p-value
Intercept (PHQ-9 – Depressive Mood Level)	16.08993	1.76530	4.17e-15 ***
Alcohol (AUDIT Score)	0.01159	0.11230	0.918
Gender (Female/Male)	-7.58067	4.61941	0.104
Interaction Alcohol:Gender	0.34169	0.27627	0.219

Table 2

PHQ-9 and AUDIT Mean and Standard Deviation (SD)

PHQ-9 (Patient Health Questionnaire)		AUDIT (Alcohol Use Disorders Identification Test)	
Female Mean:	Male Mean:	Female Mean:	Male Mean:
16.27	14.25	15.13	16.25

Female SD:	Male SD:	Female SD:	Male SD:
4.89	3.51	4.28	4.84

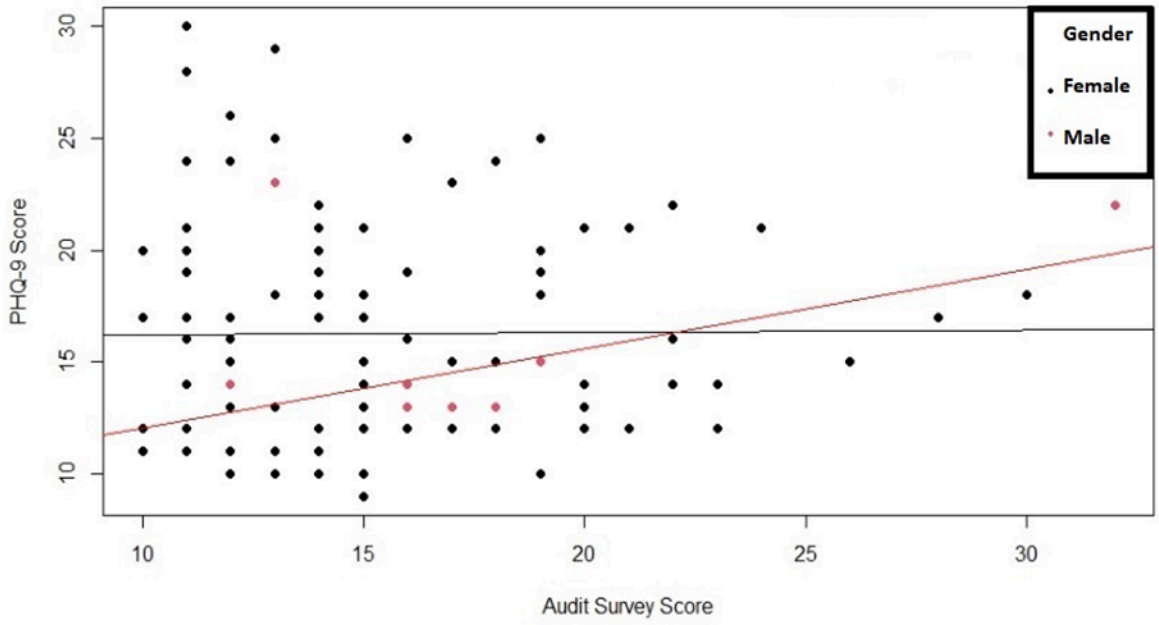

Figure 1. Multiple Regression Interaction scatter plot graph.